PROJECT DOCUMENTATION

IICET Talent & Corporate Training Portal

1. Introduction

1.1 Project Introduction

IICET Talent & Corporate Training Portal is a web-based enterprise application developed to provide a centralized digital platform for managing the complete lifecycle of industrial talent development, candidate training cohorts, and institutional operations. With technical education and corporate training playing a crucial role in workforce skill development, modern academies require an efficient mechanism to manage candidate onboarding, daily attendance logs, curriculum progress, task evaluations, and financial fee records.

Traditional institutional administration often relies on manual paper registers, disconnected desktop spreadsheets, and informal messaging channels. These conventional methods cause significant operational bottlenecks, including attendance disputes, untracked fee installments, delayed project feedback, and fragmented candidate documentation. A unified digital governance portal brings these scattered operations into a structured, secure, and transparent environment.

The IICET Talent & Corporate Training Portal organizes institutional activities through a responsive, role-based web interface supporting four distinct stakeholders: **Director (Executive Management)**, **HR Manager**, **Technical Trainers (Mentors)**, and **Candidates (Trainees)**. The platform automates onboarding, tracks daily attendance with automated late detection, facilitates programming assignment reviews, manages leave requests, and verifies student KYC documents. Additionally, it provides double-entry fee ledger management with instant PDF receipt generation, as well as AI-powered executive summarization of daily staff activity logs.

The project follows a standard three-tier application architecture:

- **Frontend:** React.js delivers an intuitive, responsive user interface with interactive role-specific dashboards, visual performance analytics, and dynamic form workflows.
- **Backend:** Node.js with Express.js handles business logic execution, Role-Based Access Control (RBAC), JWT security verification, background cron automation, and RESTful API management.
- **Database:** SQLite / MySQL provides a structured relational data store maintaining normalized records for users, attendance, task submissions, fee ledgers, and audit logs.

The frontend communicates with the backend via secure HTTP REST APIs sending and receiving structured JSON payloads. The backend validates requests, executes institutional business rules, queries the persistence layer, and returns formatted responses for display. Overall, the system eliminates administrative paperwork, minimizes human calculation errors, and establishes complete operational transparency across the institution. Its modular design ensures easy scalability for future enhancements such as placement drives and automated certification.

1.2 Existing System

In the existing traditional approach, educational academies and industrial training institutes manage student admissions, attendance, and training operations through manual registers, disconnected Excel spreadsheets, and informal communication channels (such as WhatsApp groups or paper circulars). Candidate personal information, educational background, fee records, and KYC documents are collected physically or through disparate email threads.

Similarly, institutional administrative officers and trainers maintain batch rosters, daily attendance, class schedules, and project marks in isolated physical registers or personal desktop computers. When records are maintained in fragmented storage, updating and reviewing information requires substantial manual effort.

The traditional approach creates several severe operational difficulties:

- **Manual Candidate Record Management:** Candidate personal details, contact information, qualifications, and batch allocations are recorded manually across multiple physical logbooks. When a student updates contact information or shifts batches, physical updates are delayed or lost, resulting in outdated administrative records.
- **Attendance Inaccuracies and Time Disputes:** Trainers track attendance using manual signature sheets, which lack precise timestamps. This encourages proxy attendance, leads to disputes regarding punch-in and punch-out timings, and complicates the calculation of monthly working hours.
- **Difficulty in Managing Project Tasks and Submissions:** Technical programming assignments and source code links are submitted through informal chat threads or email attachments. Mentors struggle to track submission deadlines, maintain version histories, and provide objective grading rubrics.
- **Fee Collection and Ledger Discrepancies:** Candidates frequently pay training course fees across multiple installments (Cash, Cheque, or UPI). Recording installments across manual paper ledgers frequently causes accounting mismatches, delay in reconciliation, and inconvenience in issuing formal fee receipts.
- **Lack of Centralized Executive Visibility:** Directors and institutional leaders have no real-time dashboard or unified reporting system to monitor daily staff operations, attendance statistics, or financial summaries without requesting multiple offline reports.

1.3 Need and Scope of the System

Need of the System:

- **Centralized Candidate Records:** Provides a common, secure relational repository to maintain candidate profiles, batch allotments, technology domains, contact credentials, and enrollment statuses.
- **Automated Attendance Monitoring:** Replaces error-prone paper registers with real-time clock-in and clock-out timestamps, automated late-arrival detection, and background cron auto-checkout.
- **Structured Task and Performance Evaluation:** Facilitates structured assignment allocation, project proof submission tracking, and systematic grading rubrics, eliminating lost assignments and delayed feedback.
- **Transparent Fee and Installment Management:** Maintains an automated double-entry ledger to track fee installments, calculate remaining balances, prevent accounting errors, and instantly generate digital PDF receipts.
- **Digital KYC and Document Repository:** Streamlines candidate identity verification by enabling secure online uploads of government ID proofs and resumes for authorized verification.
- **Executive Visibility & AI Summarization:** Enables executives to access institution-wide operational metrics and leverages Google Gemini AI to synthesize multi-day staff activity logs into actionable summaries.

Scope of the System:

- **Candidate (Trainee) Scope:** Supports authentication, profile maintenance, clock-in/out attendance logging, assignment submission with GitHub/live URLs, fee installment ledger review, stamped PDF receipt download, leave applications, and curriculum schedule access.
- **Technical Trainer (Mentor) Scope:** Enables mentors to view assigned batches, create technical assignments with submission deadlines, review and score trainee project submissions, manage lecture schedules, and submit daily work activity logs for management review.
- **HR & Administrative Scope:** Supports candidate onboarding, batch allotment, recording fee installment payments, verifying uploaded KYC credentials, processing leave requests with automated attendance synchronization, and managing admission enquiries.

- **Director & Executive Management Scope:** Provides high-level institutional dashboards displaying overall enrollment numbers, fee revenue analytics, staff attendance, audit logs, and AI-generated multi-day staff performance summaries.

1.4 Organization Profile

IICET Talent & Corporate Training Portal is the enterprise application developed for **IICET Services Private Limited** (Satara, Maharashtra) to organize, monitor, and automate technical training cohorts, candidate skill development, and institutional operations through a centralized digital platform.

The organization focuses on bridging the gap between academic education and corporate industry expectations by providing structured technical training in software engineering, practical development projects, and corporate skill programs. The portal utilizes modern web technologies to streamline administrative workflows and ensure seamless collaboration between candidates, technical mentors, human resources, and executive leadership through dedicated role-specific modules.

2. Proposed System

2.1 Objectives

1. **Develop a Centralized Institutional Portal:** Unifies candidate onboarding, attendance, assignment evaluations, and fee records into a single relational database.
2. **Provide a Responsive Role-Based Interface:** Features dedicated, modern dashboards tailored for Director, HR Manager, Technical Trainer, and Candidate roles.
3. **Implement Secure Role-Based Authentication:** Enforces JSON Web Tokens (JWT), Bcrypt password encryption, and Gmail SMTP email OTP verification.
4. **Automate Attendance and Time Tracking:** Automates real-time punch-in/out logging, late detection (after 9:30 AM), and 6:00 PM cron auto-checkout.
5. **Streamline Task Delegation & Evaluation:** Provides an assignment board with GitHub/live demo link submissions and objective grading rubrics.
6. **Manage Fee Ledgers & Generate PDF Receipts:** Implements double-entry fee installment calculation with instant branded PDF receipt generation.
7. **Provide Automated Executive AI Summaries:** Integrates Google Gemini 1.5 Flash AI to synthesize multi-day staff activity logs into executive insights.

2.2 Software & Hardware Specifications

Hardware Requirements: Processor: Intel Core i3 or above | RAM: 4 GB or higher | Hard Disk: 50 GB | Display: 1366 x 768 or higher.

Software	Version	Role in Project
Operating System	Windows 10 / 11 / Linux	Development and production hosting environment
Node.js Runtime	v20.x / v22.x LTS	Executes server-side JavaScript engine and REST APIs
Express.js	^4.19.2	Backend web application framework routing API endpoints
React.js	^19.2.7	Frontend library for building reactive user interfaces
Vite	^8.1.1	High-performance frontend build tool and bundler
SQLite (node:sqlite)	Native Engine	Embedded ACID-compliant primary relational database
MySQL Server	8.0 / 8.4 LTS	Secondary enterprise relational data storage and synchronization

Software	Version	Role in Project
Google Gemini AI	1.5 Flash SDK	Generative AI engine synthesizing daily staff work logs
Nodemailer	^6.9.13	Gmail SMTP protocol dispatcher for automated emails and OTPs
Postman	Latest	Testing, debugging, and documenting REST API endpoints

3. System Diagrams

3.1 Data Flow Diagrams (DFD)

/* [PLACE DIAGRAM / SCREENSHOT HERE: Level 0 DFD - Context Diagram] */

Paste your Level 0 Context Diagram here. Entities: Candidate, Trainer, HR Manager, Director. Central Process: 0.0 IICET Talent & Corporate Training Portal.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Level 1 DFD - Administrator & HR Workflow] */

Paste your Level 1 Admin DFD here. Processes: 1.0 Auth, 1.1 Candidate Onboarding, 1.2 Fee Management, 1.3 Leave Approval, 1.4 KYC Verification. Data Stores: D1 Users, D2 Interns, D3 Fees, D4 Attendance.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Level 1 DFD - Candidate & Trainer Workflow] */

Paste your Level 1 Candidate & Trainer DFD here. Processes: 2.0 Attendance Logging, 2.1 Task Allocation, 2.2 Task Submission, 2.3 Grading.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Level 2 DFD - Subsystem Decomposition] */

Paste your Level 2 DFD here. Decomposition into Attendance Engine, Task Review Board, and Fee PDF Receipting subsystems.

3.2 Entity Relationship Diagram (ERD)

/* [PLACE DIAGRAM / SCREENSHOT HERE: Entity Relationship Diagram (ERD)] */

Paste your full database ER Diagram here showing all primary keys, foreign keys, and cardinalities.

3.3 Use Case Diagrams

/* [PLACE DIAGRAM / SCREENSHOT HERE: Use Case Diagram - Director & HR Administration] */

Paste your Director & HR Use Case Diagram here.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Use Case Diagram - Candidate & Technical Trainer] */

Paste your Candidate & Trainer Use Case Diagram here.

3.4 Activity & Sequence Diagrams

/* [PLACE DIAGRAM / SCREENSHOT HERE: Activity Diagram - Administrative & Candidate Flow] */

Paste your Activity Diagram showing operational workflows from login to task evaluation and receipt generation.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Sequence Diagram - System Messaging Flow] */

Paste your Sequence Diagram showing interaction between Web Browser, Express REST API, and SQLite/MySQL Database.

4. System Design & Data Dictionary

The Data Dictionary documents the relational schema definitions, field data types, nullability, constraints, and descriptions across the 12 normalized database tables:

1. Users Table (Core Authentication & Role Management)

Field Name	Data Type	Constraints	Description
id	INTEGER	PK, AUTO_INCREMENT	Unique user ID
name	VARCHAR(100)	NOT NULL	Full legal name
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email address
role	VARCHAR(20)	NOT NULL	Role ('Director', 'HR', 'Mentor', 'Intern')
department	VARCHAR(100)	NOT NULL	Assigned department ('Director', 'Human Resources', 'Employee', 'Intern')
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
mobile_number	VARCHAR(20)	NULLABLE	Phone contact number
is_online	TINYINT(1)	DEFAULT 0	Online session status (1=Active, 0=Offline)

2. Interns Table (Candidate Profile Extension)

Field Name	Data Type	Constraints	Description
id	INTEGER	PK, AUTO_INCREMENT	Unique intern profile ID
user_id	INTEGER	FK (users.id), UNIQUE	Reference to core user account
batch	VARCHAR(50)	NOT NULL	Cohort designation (e.g., 'Batch 2026-A')
training_time	VARCHAR(50)	NOT NULL	Scheduled training hours
mentor_id	INTEGER	FK (users.id)	Assigned trainer/mentor ID

Field Name	Data Type	Constraints	Description
technology	VARCHAR(100)	NOT NULL	Track ('Full Stack Development', 'Data Science')
status	VARCHAR(20)	DEFAULT 'Active'	Training status ('Active', 'Completed')

3. Attendance Table (Daily Punch Clock & Cron)

Field Name	Data Type	Constraints	Description
id	INTEGER	PK, AUTO_INCREMENT	Unique attendance record ID
intern_id	INTEGER	FK (interns.id)	Reference to intern profile
date	DATE	NOT NULL	Session date (YYYY-MM-DD)
check_in_time	VARCHAR(20)	NULLABLE	Punch-in timestamp (HH:MM:SS)
check_out_time	VARCHAR(20)	NULLABLE	Punch-out timestamp (HH:MM:SS)
status	VARCHAR(20)	NOT NULL	Status ('Present', 'Late', 'Half Day', 'Absent')
working_hours	DECIMAL(4,2)	DEFAULT 0.00	Total computed working hours

4. Tasks & Task Updates Tables (Assignment Kanban & Grading)

Field Name	Data Type	Constraints	Description
id	INTEGER	PK, AUTO_INCREMENT	Unique task assignment ID
title	VARCHAR(200)	NOT NULL	Task title / assignment heading
assigned_by	INTEGER	FK (users.id)	Trainer or Director user ID
assigned_to	INTEGER	FK (interns.id)	Assigned candidate ID
priority	VARCHAR(20)	DEFAULT 'Medium'	Urgency ('High', 'Medium', 'Low')
due_date	DATE	NULLABLE	Submission deadline date
status	VARCHAR(20)	DEFAULT 'Pending'	Status ('Pending', 'Submitted', 'Approved')

5. Fees & Payments Tables (Double-Entry Ledgers & PDF Receipts)

Field Name	Data Type	Constraints	Description
id	INTEGER	PK, AUTO_INCREMENT	Unique transaction ID
intern_id	INTEGER	FK (interns.id)	Reference to paying candidate
total_amount	DECIMAL(10,2)	NOT NULL	Total tuition fee agreed
paid_amount	DECIMAL(10,2)	DEFAULT 0.00	Total cumulative paid
remaining_amount	DECIMAL(10,2)	NOT NULL	Outstanding balance dues
receipt_number	VARCHAR(50)	UNIQUE, NOT NULL	Official receipt serial number
payment_mode	VARCHAR(50)	NOT NULL	Payment mode ('Cash', 'UPI', 'Net Banking')

4.2 Input and Output Interface Designs

/* [PLACE DIAGRAM / SCREENSHOT HERE: Portal Login Screen] */

Unified portal authentication interface supporting role selection (Director, HR, Trainer, Intern), email address input, show/hide password toggle, and direct authentication.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Executive Director Dashboard] */

High-level administrative console displaying total active trainees, staff attendance counts, monthly revenue collections, AI-synthesized daily activity highlights, and institution-wide metrics.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Candidate Dashboard] */

Student-facing portal featuring attendance check-in/out widgets, assigned project tasks with due dates, recent grades, remaining fee balance summaries, and lecture schedules.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Attendance Management Console] */

Multi-role biometric-style attendance tracker showing daily logs, clock-in/out timestamps, automated late detection flags, working hour computation, and monthly CSV export.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Task Assignment & Kanban Review Board] */

Mentorship console enabling technical trainers to create assignments, define problem statements, monitor submissions, review GitHub/demo links, and submit grades with feedback.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Fee Ledger & Stamped PDF Receipt Screen] */

Double-entry accounting module showing candidate installment history, payment modes (Cash, UPI, Card), and client-side instant generation of official PDF receipts with corporate branding.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Digital KYC & Document Repository] */

Document verification interface allowing candidates to upload identity proofs and resumes, with HR verification workflows (Uploaded, Approved, Rejected).

/* [PLACE DIAGRAM / SCREENSHOT HERE: Automated Gemini AI Daily Log Summarizer] */

Executive intelligence console synthesizing multiple daily trainer and HR work logs into structured bullet points for leadership decision-making.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Help Desk & Multi-Tier Ticketing Console] */

Support ticketing interface enabling candidates to raise queries to specific trainers or HR, complete with priority indicators and resolution response threads.

/* [PLACE DIAGRAM / SCREENSHOT HERE: Candidate Profile & Department View] */

Institutional profile card displaying candidate credentials, assigned department ('Intern'), batch timing, academic qualifications, and verified contact details.

5. System Modules

5.1 Candidate (Intern) Module

Provides registered trainees with an integrated interface for daily clock-in/out attendance, task submission with GitHub/live demo links, fee balance tracking, stamped PDF receipt download, and KYC document uploads.

5.2 Role-Based Authentication & Security Module

Enforces Role-Based Access Control (RBAC), JSON Web Token (JWT) session security, Bcrypt password hashing, and real-time Gmail SMTP email OTP verification.

5.3 Multi-Role Attendance Engine & Cron Daemon

Tracks precise punch-in/out timestamps, applies automated late-arrival flags (after 9:30 AM), and runs a background cron engine that automatically checks out unclosed sessions at 6:00 PM.

5.4 Task Delegation & Review Board Module

Allows technical mentors to create programming assignments with deadlines and score trainee submissions using objective grading rubrics (0 - 100 marks and grades A+ through F).

5.5 Fee Ledger, Double-Entry Tracking & PDF Receipts

Maintains candidate financial accounts, updates remaining balances in atomic transactions, and generates client-side official signed PDF receipts with company credentials.

5.6 Digital KYC & Document Verification Module

Centralizes storage of Aadhar cards, college IDs, resumes, and offer letters with administrative verification workflows ('Uploaded', 'Verified', 'Rejected').

5.7 Daily Work Logs & Google Gemini AI Summarizer

Synthesizes multi-day staff activity logs into structured executive bullet points using Google Gemini 1.5 Flash AI to give directors clear institutional intelligence.

5.8 Leave Requests & Attendance Synchronization

Facilitates online leave submissions with review approval workflows that automatically synchronize with daily attendance to prevent incorrect absence marks.

5.9 Help Desk & Multi-Tier Ticketing Module

Enables candidates and mentors to log queries directly to HR or the Director, complete with priority indicators and email notification alerts.

5.10 Institutional Auditing & Reporting Module

Generates comprehensive audit spreadsheets across attendance, tasks, and financial ledgers with CSV export capabilities for academic and accreditation audits.

6. Advantages of the System

- **Centralized Institutional Records:** Consolidates candidate records, batch rosters, staff attendance, task evaluations, and fee transactions into a single relational database, eliminating conflicting spreadsheets.
- **High Attendance Accuracy:** Eliminates proxy sign-ins and manual calculation of working hours through precise timestamps, automated late-arrival flags, and background cron auto-checkout.
- **Financial Transparency & Error Prevention:** Enforces mathematical consistency across fee installments, preventing double entry or overpayments while providing instant, tamper-resistant PDF receipts.
- **Structured Performance Tracking:** Accelerates feedback loops between mentors and candidates through real-time task allocation, repository link submissions, and structured grading rubrics.
- **Streamlined KYC Compliance:** Automates identity proof uploads, preventing lost paper files and expediting candidate background checks.
- **AI-Powered Executive Intelligence:** Google Gemini Generative AI empowers executive directors with summarized insights into institutional productivity without requiring manual log collation.

7. Future Scope

- **Dedicated Mobile Application:** Developing cross-platform mobile apps using Flutter or React Native to enable candidates and trainers to punch attendance, view grades, and receive push notifications directly on mobile phones.
- **Biometric & Geo-Fencing Integration:** Integrating biometric fingerprint scanners or GPS-based geofencing to verify physical presence within campus premises during attendance clock-in.
- **Automated Online Payment Gateway:** Connecting payment gateways such as Razorpay or Stripe to allow students to pay course fee installments directly via credit card, debit card, or net banking with instant webhooks.
- **Corporate Placement & Hiring Portal:** Implementing dedicated recruitment portal modules allowing corporate hiring partners to browse verified candidate resumes, view coding project scores, and schedule campus interviews.
- **Blockchain / QR-Verified Certification:** Automated generation of digitally signed, QR-code-verifiable course completion certificates based on minimum attendance and task completion thresholds.

8. Conclusion

The **IICET Talent & Corporate Training Portal** is an enterprise-grade full-stack web application designed to centralize and automate the end-to-end administration of corporate training cohorts, candidate lifecycle management, and institutional governance.

The project utilizes React.js (Vite) to deliver a modern, responsive user interface across four role-based portals (Director, HR, Trainer, Candidate). The backend is engineered with Node.js and Express.js, providing secure RESTful API endpoints, Role-Based Access Control (RBAC), JWT authentication, background cron automations, and Google Gemini 1.5 Flash AI integration. High-integrity data persistence is achieved through a normalized relational

architecture supporting both embedded SQLite and enterprise MySQL engines.

By automating attendance monitoring, assignment evaluations, fee ledger reconciliations, document verification, and administrative reporting, the system eliminates paperwork, minimizes human calculation errors, and establishes complete operational transparency across all institutional departments. Its modular structure establishes a strong foundation for future advancements, demonstrating how modern web technologies and artificial intelligence can transform institutional management.

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